**RTCA, Inc.**

1828 L Street, NW, Suite 805

Washington, DC 20036

# RTCA/DO-160G

# Environmental Conditions and Test Procedures for Airborne Equipment

## Section 10

## Waterproofness

**Important Notice**

Information contained in these sections is pertinent to all test procedures described in the other sections of this document. Further, Appendix A is applicable for identifying environmental tests performed.

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# 10.0 Waterproofness

## 10.1 Purpose of the Test

These tests determine whether the equipment can withstand the effects of liquid water being sprayed or falling on the equipment or the effects of condensation.

These tests are not intended to verify performance of hermetically sealed equipment. Therefore, hermetically sealed equipment may be considered to have met all waterproofness requirements without further testing. Equipment shall be considered hermetically sealed when the seal is permanent and airtight.

## 10.2 Equipment Categories

Category Y

Equipment that is installed in locations where it is subjected to condensing water in the course of normal aircraft operations is identified as Category Y. For equipment intended for installation in such locations, the condensing water proof test procedure applies and the equipment is identified as Category Y.

Category W

Equipment that is installed in locations where it is subjected to falling water (generally the result of condensation) in the course of normal aircraft operations is identified as Category W. For equipment intended for installation in such locations, the drip proof test procedure applies and the equipment is identified as Category W.

Category R

Equipment installed in locations where it may be subjected to a driving rain or where water may be sprayed on it from any angle is identified as Category R. For equipment intended for installation in such locations, the spray proof test procedure applies. Equipment that has passed the Category R requirements may be considered to meet the Category W requirement without further testing.

Category S

Equipment installed in locations where it may be subjected to the forces of a heavy stream of fluid such as would be encountered in aircraft de-icing, washing or cleaning operations is identified as Category S. For equipment intended for installation in such locations the continuous stream proof procedure applies. Water is used in this test to simulate the actual fluid forces. Equipment that has passed the Category S requirements may be considered to meet the Category W requirements without further testing.

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## 10.3 Test Procedures

### 10.3.1 Condensing Water Proof Test

Two temperature chambers will be used for this test. Chamber 1 will be set to -10°C, while chamber 2 will be set to 40°C and 85% RH. Mount the equipment under test according to the manufacturer's specifications with all connectors and fittings engaged into chamber 1. With the equipment not operating let the equipment stabilize for a minimum of three hours. Chamber 2 shall be stabilized at 40°C and 85% humidity during this same time. After the three-hour stabilization, transfer the equipment to chamber two within a five minute transition time. Mount the equipment according to the manufacturer's specifications with all connectors and fittings, including ventilation, engaged into chamber 2 within a maximum delay of 5 minutes. Turn the equipment on and let it operate for 10 minutes (+1 minute, - 0 minutes). Following the 10 minute operating period DETERMINE COMPLIANCE WITH APPLICABLE EQUIPMENT PERFORMANCE STANDARDS.

### 10.3.2 Drip Proof Test.

Prior to the start of the Drip Proof test, with the equipment under test not operating, stabilize the equipment to a temperature of at least 10 degrees Celsius above the temperature of the water to be used for the drip test. Place equipment according to the manufacturer's specifications with all connectors and fittings engaged. With the equipment not operating, subject it to water falling at a uniform rate from a minimum height of one meter above the top surface of the equipment under test for a minimum of 15 minutes. The test equipment shall emit a volume of water greater than 140 $l/m^2/hr$ dripping from a dispenser with 0.33 mm nominal diameter drip holes on a 25 mm pattern as shown in Figure 10-1. Note: To meet the $140 l/m^2/hr$ or greater volume of water. A pre-calibration and post calibration of the volume of water emitted from the drip table shall be conducted and the level in the drip table shall be noted. The drip hole pattern shall be sufficiently large to meet or exceed the horizontal cross sectional area of the equipment under test when installed in its normal position. At the conclusion of the test DETERMINE COMPLIANCE WITH APPLICABLE EQUIPMENT PERFORMANCE STANDARDS.

### 10.3.3 Spray Proof Test

Mount the equipment according to the manufacturer's specification with all connectors and fittings engaged. With the equipment operating, subject it to a shower of water from a shower head nozzle as depicted in Figure 10-2. The water shall be directed perpendicular to the most vulnerable area(s) of the equipment as stated in the applicable equipment performance standards.

Each of the areas under test shall be subjected to the spray for a minimum of 15 minutes. If desired, the test may be applied simultaneously to more than one area at a time by using an appropriate number of showerheads. The showerhead shall be located not more than 2.5 m from the area under test and shall emit a volume of water greater than 450 liters per hour. At

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the conclusion of the test DETERMINE COMPLIANCE WITH APPLICABLE EQUIPMENT PERFORMANCE STANDARDS.

### 10.3.4 Continuous Stream Proof Test

This test is used to supplement the fluids susceptibility test in Section 11.0. Susceptible materials such as gaskets shall be subjected to the appropriate tests of Section 11.0 prior to the performance of this test. This test shall be performed with water at a temperature of 50 degrees C.

Mount the equipment according to the manufacturer's installation instructions in a manner that simulates the aircraft installation. Connectors or other fittings shall be connected as in normal operation. The equipment need not be operated during this test.

Subject the equipment, particularly in areas where parts are mated with a resilient gasket, to a continuous stream of water on all sides for a minimum five minutes on each side. The stream of water shall be of sufficient pressure to produce, through a 6.4-mm diameter nozzle, at least a six-meter vertical stream of water. The equipment shall be subjected to this stream of water from a distance of one to two meters. At the conclusion of the test DETERMINE COMPLIANCE WITH APPLICABLE EQUIPMENT PERFORMANCE STANDARDS.

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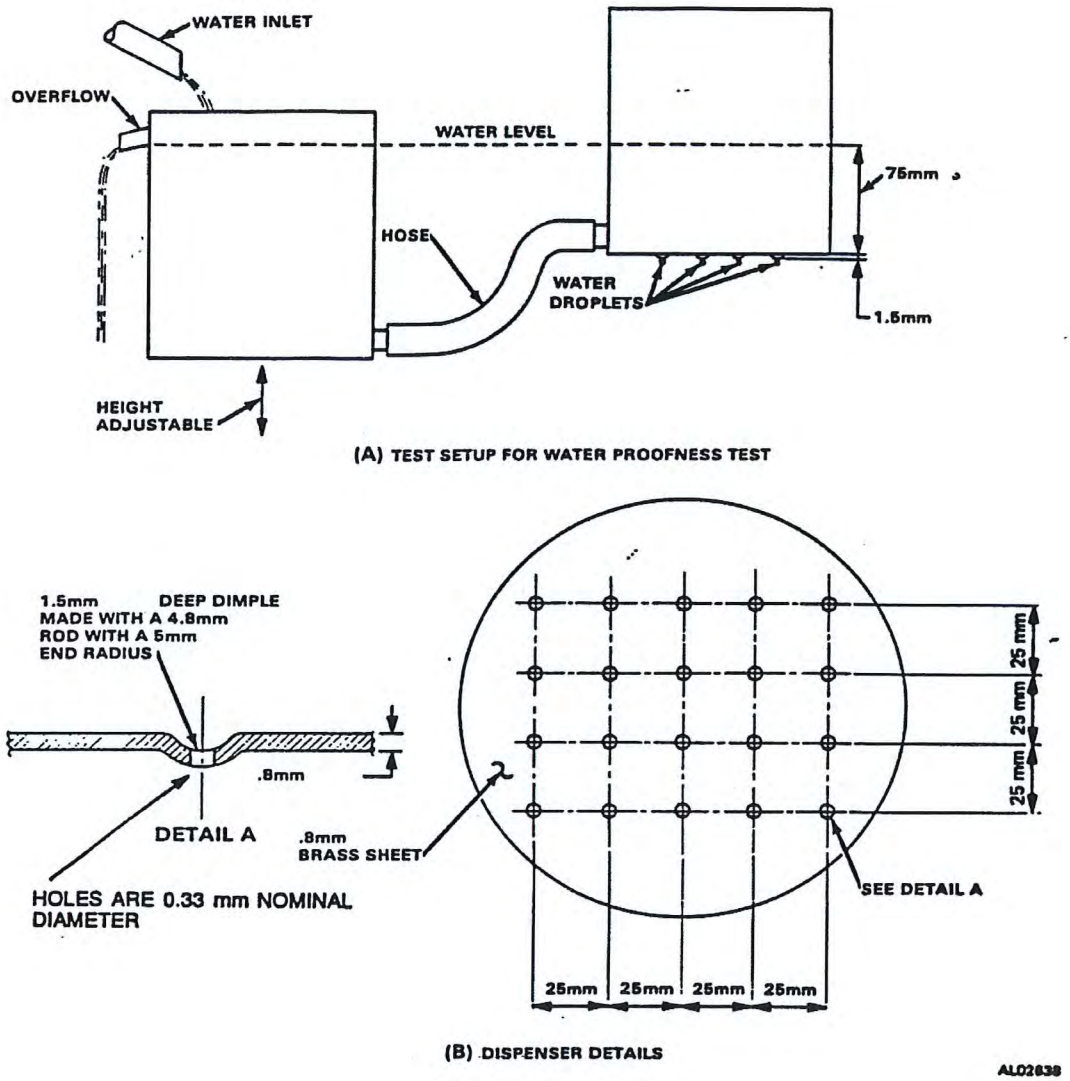


NOTE: Container size and number of holes as required to meet the flow rate requirement of paragraph 10.3.1.

**FIGURE 10-1 DRIP PROOF TEST DETAILS**

Note: To meet the 140l/m$^2$/hr or greater volume of water. A pre-calibration and post calibration of the volume of water emitted from the drip table shall be conducted and the level in the drip table shall be noted.

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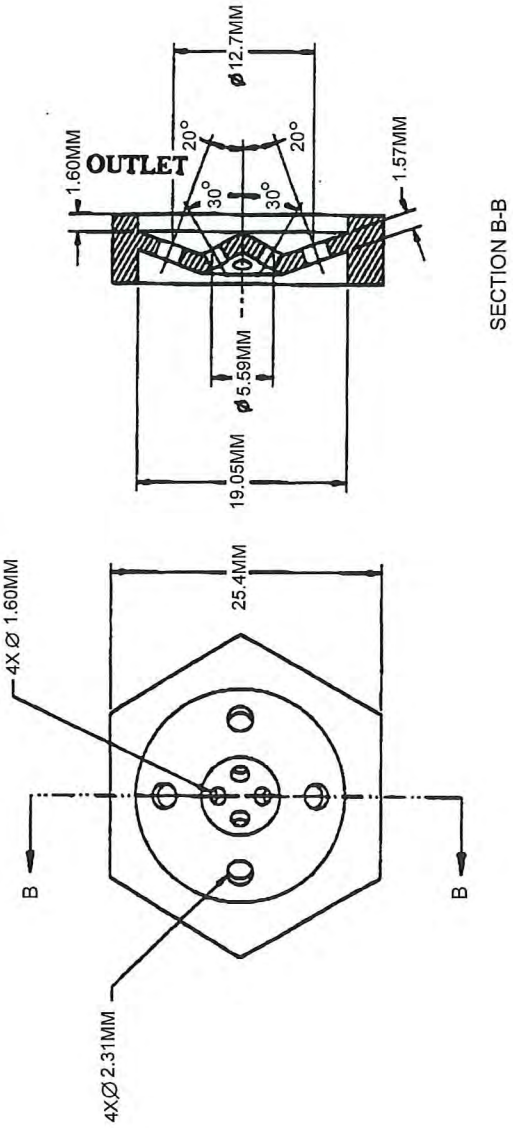


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**FIGURE 10-2 SHOWER HEAD DETAILS**

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